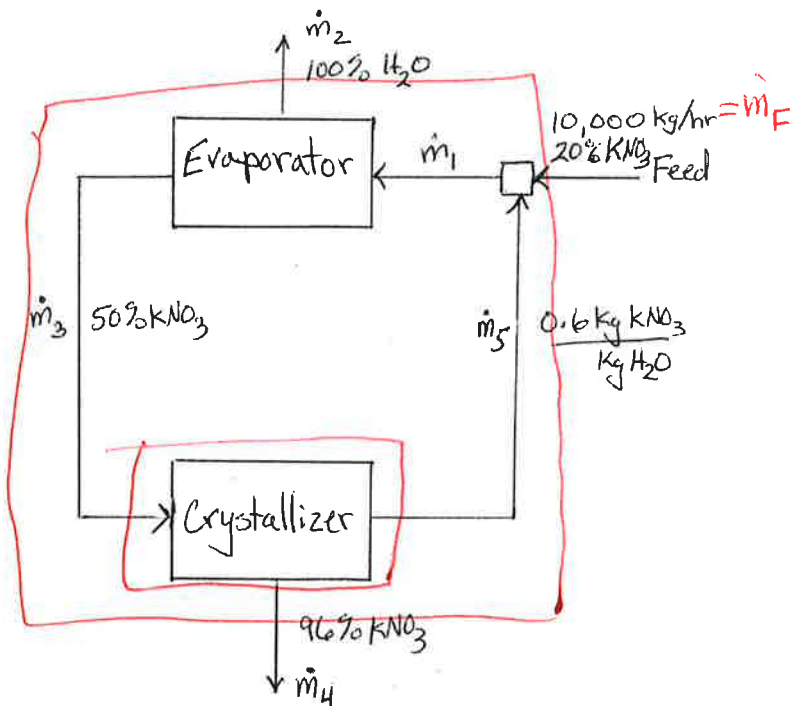


To receive full credit, you must show all of your work.

- (1) (25 pts.) Potassium nitrate (KNO_3) is separated as suspended solids from water (H_2O) by evaporation and then crystallization. A steady, continuous process is shown in the diagram. Calculate the five unknown mass flow rates $\dot{m}_1$, $\dot{m}_2$, $\dot{m}_3$, $\dot{m}_4$, and $\dot{m}_5$ in kg/hr. The feed enters at 10,000 kg/hr and 20 wt% KNO_3 . Pure water leaves the top of the evaporator and a second stream of 50 wt% KNO_3 solution also leaves the evaporator. The crystallizer makes crystals that contain only 4 wt% water and a second saturated stream with 0.6 kg KNO_3 /kg H_2O which is recycled with the feed. Clearly indicate where you define your system boundaries. These should be carefully chosen to make the problem easier to solve.



Overall Balance

Total mass: $\dot{m}_{\text{FEED}} = \dot{m}_2 + \dot{m}_4$

KNO_3 : $0.2\dot{m}_F = 0.96\dot{m}_4 \Rightarrow \dot{m}_4 = \frac{(0.2)(10,000 \text{ kg/hr})}{0.96}$
 $= 2083 \frac{\text{kg}}{\text{hr}}$

$\dot{m}_2 = \dot{m}_F - \dot{m}_4 = (10,000 - 2083) \frac{\text{kg}}{\text{hr}}$
 $= 7917 \frac{\text{kg}}{\text{hr}}$

Crystallizer balance

Total mass: $\dot{m}_3 = \dot{m}_4 + \dot{m}_5$

$w_{\text{KNO}_3} = \frac{0.6}{1.6} = 0.375$

KNO_3 : $0.5\dot{m}_3 = 0.96\dot{m}_4 + 0.375\dot{m}_5$

$0.5(\dot{m}_4 + \dot{m}_5) = 0.96\dot{m}_4 + 0.375\dot{m}_5$

$0.125\dot{m}_5 = 0.46\dot{m}_4$

$\dot{m}_5 = \frac{0.46(2083)}{0.125} = 7667 \frac{\text{kg}}{\text{hr}}$

$$\begin{aligned}\dot{m}_3 &= \dot{m}_4 + \dot{m}_5 \\ &= 2083 + 7667 \\ &= 9753\end{aligned}$$

Mixer Balance

$$\dot{m}_F + \dot{m}_5 = \dot{m}_1$$

$$10,000 + 7667 = 17,667 = \dot{m}_1$$

$$\dot{m}_1 = \underline{17,667} \text{ kg/hr,}$$

$$\dot{m}_2 = \underline{7917} \text{ kg/hr,}$$

$$\dot{m}_3 = \underline{9753} \text{ kg/hr,}$$

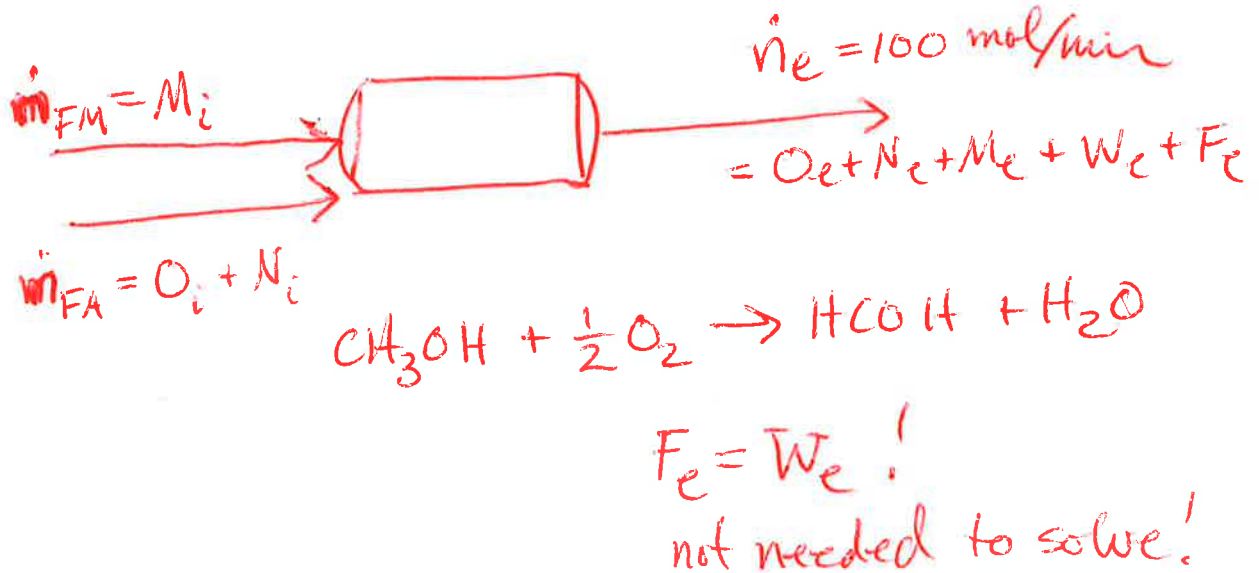
$$\dot{m}_4 = \underline{2083} \text{ kg/hr,}$$

$$\dot{m}_5 = \underline{7667} \text{ kg/hr}$$

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(2) (45 pts.) Formaldehyde (HCOH) is produced by oxidation of methanol (CH₃OH). A reactor is fed a stream of pure methanol $\dot{n}_{FM}$ and a stream of air (21 mol% O₂ and 79 mol% N₂) $\dot{n}_{FA}$. A single product stream of 100 kmol/min leaves the reactor containing formaldehyde, water (H₂O), oxygen, nitrogen, and methanol. Chemical analysis determined that the product stream is 3.4 mol% oxygen and 42.7 mol% nitrogen.

- (a) Draw and label a block flow diagram of the process described above. Represent the species molar flow rates as N_i, O_i, and M_i for inlet nitrogen, oxygen, and methanol and N_e, O_e, M_e, W_e, and F_e for the exit nitrogen, oxygen, methanol, water, and formaldehyde. Notice that $\dot{n}_{FM} = M_i$ and $\dot{n}_{FA} = N_i + O_i$. Write a balanced chemical reaction equation for the process. What can you conclude about the mole fractions of the two reaction products [H₂O and HCOH] in the product stream? (10 pts.)



- (b) Do a degree of freedom analysis following the method on page 105 of the textbook by Murphy. Fill in the table below giving a brief explanation for your entries. Compute the DOF. (5 pts.)

	Number	Explanation
4a. Stream Variables	8	8 stream comp's
4b. System Variable	1	1 Rx
Total Independent Variables	9	
5a. Specified Flows	1	$\dot{n}_e = 100 \text{ mol/min}$
5b. Specified Stream Compositions	3	0.7 N ₂ in 0.427 N ₂ out 0.034 O ₂ out
5c. Specified System Performances	0	
5d. Material Balance Equations	5	5 component O, N, M, W, F
Total Independent Equations	9	
DOF	0	

(c) Using atomic balances, calculate the molar flow rates for the two feed streams in kmol/min for the methanol $\dot{n}_{FM}$ and the $\dot{n}_{FA}$. Also calculate the mole fractions of formaldehyde y_F , methanol y_M , and water y_W in the product stream. (30 pts.)

$$C: M_i = M_e + F_e \quad E_{4H} - 4\bar{W}_e \Rightarrow F_e = \bar{W}_e$$

$$H: 4M_i = 4M_e + 2F_e + 2\bar{W}_e$$

$$N: 2N_i = 2N_e = 2(42.7) \Rightarrow N_i = \underline{42.7 \frac{\text{mol}}{\text{min}}}$$

$$O: 2O_i + M_i = 2O_e + (M_e + \bar{W}_e + F_e) \quad \left\{ \begin{array}{l} O_i = \frac{0.21}{0.79} N_i = 11.35 \\ \dot{m}_{FA} = 42.7 + 11.35 = 54.05 \end{array} \right.$$

$$\text{constrain } \dot{m}_e = 100 \frac{\text{mol}}{\text{min}} = O_e + N_e + (M_e + \bar{W}_e + F_e) \quad \left\{ \begin{array}{l} \Rightarrow (M_e + N_e + O_e) = 100 - O_e - N_e = 100 - 42.7 - 11.35 = 53.9 \\ = 53.9 \end{array} \right.$$

$$M_i = 2O_e + (53.9) - 2O_i = 2(3.4) + 53.9 - 2(11.35) = 38.00 \frac{\text{mol}}{\text{min}}$$

Subtract $E_{4H} - E_{4C}$

$$2O_i = 2O_e + \bar{W}_e \Rightarrow \bar{W}_e = 2(O_i - O_e) = 2(11.35 - 3.4) = 15.9 = \bar{W}_e = F_e$$

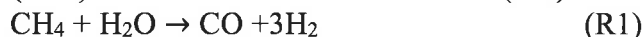
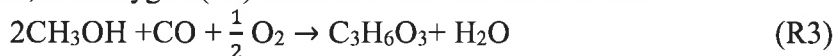
$$M_e = 53.9 -$$

$$C \Rightarrow M_e = M_i - F_e = 38 - 15.9 = 22.1$$

$$\dot{n}_{FM} = \underline{38.00} \text{ kmol CH}_3\text{OH /min}, \quad \dot{n}_{FA} = \underline{54.05} \text{ kmol air /min}, \quad y_F = \underline{0.159} \text{ mol HCOC/mol},$$

$$y_M = \underline{0.221} \text{ mol CH}_3\text{OH /mol}, \quad y_W = \underline{0.159} \text{ mol H}_2\text{O /mol}$$

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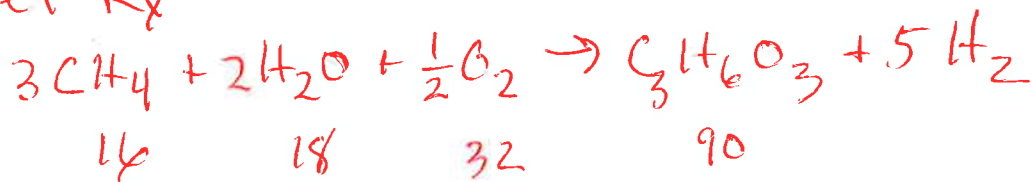
(3) (30 pts.) A new synthesis mechanism for dimethyl carbonate (DMC, $C_3H_6O_3$) is as follows:Methane (CH_4) and water (H_2O) react to form carbon monoxide (CO) and hydrogen (H_2):CO and H_2 react to form methanol (CH_3OH): CH_3OH , CO, and oxygen (O_2) react to form DMC and water.(a) Use a generation-consumption analysis to synthesize a reaction pathway to DMC based on the three reaction above with no net generation or consumption of CO or CH_3OH . Hydrogen is allowed as a byproduct. Show your calculations and write the net chemical equation for the process. (20 pts.)

	R1	R2	R3	Net Rxn
16 CH_4	-1 (-3)	0	0	-3
18 H_2O	-1 (-3)	0	(+1)	-2
2 H_2	+3 (+9)	-2 (-4)	0	+5
38 CO	+1 (+3)	-1 (-2)	(-1)	0
CH_3OH	0	+1 (+2)	(-2)	0
32 O_2	0	0	(-1/2)	-1/2
90 $C_3H_6O_3$	0	0	(+1)	+1

$$\begin{aligned} CO \quad x_1 - x_2 - x_3 &= 0 \\ CH_3OH \quad x_2 - 2x_3 &= 0 \\ \hline &+3x_3 \end{aligned}$$

$$\begin{aligned} x_1 &= 1 \\ x_3 &= \frac{1}{3} \\ x_2 &= \frac{2}{3} \end{aligned}$$

Net Rx



$$\begin{aligned} x_3 &= 1 \\ x_1 - x_2 &= 1 \\ x_2 &= 2 \\ x_1 &= 3 \end{aligned}$$

$$(b) \quad \frac{(1)(90)}{(3)(16) + (2)(18) + (\frac{1}{2})(32)} = \frac{90}{48 + 36 + 16} = \frac{90}{100} = 0.9$$

- (b) Calculate the fractional economy of the overall reaction. (Assume the atomic weights of carbon, hydrogen, and oxygen are: $M_C = 12$ g/mole, $M_H = 1$ g/mole, and $M_O = 16$ g/mole, respectively.) (10 pts.)